

Student 76

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In a grammar, **terminals** and **non-terminals** are the fundamental symbols used to formally describe the syntax of a language. Together, they allow the grammar to define which strings belong to the language and how those strings are structured.

Terminals are the basic, indivisible symbols of the language that appear in the final strings generated by the grammar. They represent the actual characters or tokens that a programmer writes in source code and that a compiler ultimately recognizes. Terminals cannot be replaced by any other symbols during the derivation process. For example, in a programming language grammar, keywords such as `if`, `while`, or `int`, operators like `+`, `-`, `*`, punctuation symbols such as `;`, `(`, `)`, and even identifiers and numeric constants are treated as terminals. When a grammar generates a valid sentence, the final result consists only of terminal symbols.

Non-terminals, on the other hand, are symbolic placeholders used to define the structure and rules of the language. They represent syntactic categories or constructs and are used on the left-hand side of production rules. Unlike terminals, non-terminals do not appear in the final generated string; instead, they are repeatedly replaced by other non-terminals or terminals until only terminals remain. For example, symbols such as `Expression`, `Statement`, or `Declaration` are common non-terminals in programming language grammars. These symbols help describe how smaller components of the language combine to form larger constructs.

To illustrate the difference, consider a simple grammar rule such as

`Expression → Expression + Term | Term.`

Here, `Expression` and `Term` are non-terminals because they describe grammatical structures, while the symbol `+` is a terminal because it appears directly in the final expression. If the grammar generates the string `a + b`, the identifiers `a` and `b` and the operator `+` are terminals, while the non-terminals are used only during the derivation process.

Overall, terminals define the actual vocabulary of the language, while non-terminals define its grammatical organization. Both are essential, as terminals give meaning to the

language, and non-terminals provide the rules that ensure the language is syntactically correct.